

Introduction to Topology

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1 Introduction

Topology studies mathematical spaces where quantitative features such as distance, angle, volume, and area are ignored, and qualitative features such as openness, continuity, connectedness, compactness, and holes are focused on instead. Given a structureless set A , the minimum amount of data you can append to A such that it properly becomes a *space* with these most basic qualities is exactly a topology.

These notes introduce general topology and some topics of algebraic topology suitable for a strong undergraduate. Some exposure to topological topics from, e.g., an analysis course is helpful. We begin with metric spaces, where a notion of distance is still present; the open sets of a metric space are the motivating example of the abstract topologies defined afterwards.

2 Metric spaces

Definition 2.1 (metric spaces). Given a set X , a function $d : X \times X \rightarrow [0, \infty)$ is a *metric* if these hold for all $x, y, z \in X$:

- (a) $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles);
- (b) $d(x, y) = d(y, x)$ (symmetry);
- (c) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

A set together with a metric (X, d) is called a *metric space*.

Example 2.1. $\mathbb{R}$ has the *standard metric* $d(x, y) = |x - y|$, and more generally $\mathbb{R}^n$ has the Euclidean metric $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$. Any set X also admits the *discrete metric*

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y, \end{cases}$$

in which any two distinct points are distance 1 apart.